

Review

Data-Driven Decision Support in SaaS Cloud-Based Service Models

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Abstract: Software as a service (SaaS) is a major service model for delivering software to end users through the cloud. SaaS platforms provide their users with cost-efficient, flexible and scalable services that can be available on demand, anytime, and anywhere. Moreover, SaaS empowers software providers to establish recurring revenue and create profitable businesses. However, SaaS can endure high customer turnover due to reasons such as serving a wide range of customers, intense competition and rapid evolution of technology. Maintaining a regular customer base and keeping users engaged is crucial for the survival of a SaaS business. Thus, it is crucial for SaaS providers to identify both the reasons behind users' engagement and churn of their app towards taking proper actions to retain them in the long term. SaaS data regarding user behavior, subscriptions and system performance can be utilized for deriving insights and identifying patterns to support decision-making for SaaS providers. To this end, the aim of this survey is to review research in data-driven decision support systems in SaaS, identifying current gaps and challenges and highlighting directions for future improvements towards the development of more efficient and intelligent systems.

Keywords: software as a service; churn prediction; user engagement; machine learning; usage mining



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1. Introduction

Cloud computing is the dominant model for providing software and technology infrastructures nowadays. This approach enables access to computing resources and applications without the need for hardware setup and maintenance. The main cloud service models are infrastructure as a service (IaaS), platform as a service (PaaS) and software as a service (SaaS) [1].

The IaaS model provides infrastructure resources through the cloud such as storage and computational power. Users do not have to update and maintain the infrastructure, yet they are in charge of setting up operating systems, software and data [2]. On the contrary, the PaaS model also provides the software resources for creating software in the cloud. Users do not maintain the software development environment, but they have to write their own code. Finally, the SaaS model provides ready-to-use software solutions [3], meaning that the users can start using the software without installation. Therefore, IaaS and PaaS users are mainly developers and IT professionals, while SaaS users can vary depending on the solution provided by the SaaS. For example, a SaaS that provides invoicing software may have users ranging from various professions, e.g., engineers, lawyers, etc. The SaaS model is the leading cloud computing model for delivering software nowadays since it

enables software distribution without the need for installation and hardware setup [4]. Thus, users can use them easily and quickly, while generally paying by subscription instead of a large amount upfront.

The SaaS model enables software providers to create profitable businesses allowing software distribution through the cloud to many users, simultaneously [5]. Moreover, the subscription-based business model provides opportunities for stable income streams. Thus, the success of a SaaS model as a business heavily depends on whether it maintains a core user base. When users experience meaningful benefits from a software solution, they use it regularly and for a long time. Engaged users are the superpower of SaaS since they tend to use it in the long term and with minimum support, while they could possibly suggest it to other users. The cost of obtaining new users is generally much higher than retaining the ones that already exist [6]. Thus, SaaS businesses strive to increase user engagement and reduce user churn, i.e., the percentage of users that cancel their subscriptions.

Recent industry reports point to several growing challenges that make strong, data-driven decision support systems more important than ever for SaaS companies. One major issue is the high rate of customer churn, shortly after customers sign up, which is commonly observed with new artificial intelligence (AI)-powered tools (<https://www.paddle.com/blog/saas-market-report-q1-2025>, assessed on 1 June 2025). This is particularly a problem in the B2C market, where user behavior changes quickly and is hard to predict. At the same time, many SaaS businesses do not pay enough attention to how they adjust their prices, even though pricing can have a big impact on user retention (<https://www.omnibus.com/blog/saas-industry-report-2024>, assessed on 1 June 2025). Another challenge is the rise of multi-cloud and hybrid systems, where companies run their software across several cloud platforms. This makes it harder to gather and analyze all of their data in one place (<https://www.fortunebusinessinsights.com/software-as-a-service-saas-market-102222>, assessed on 1 June 2025). The growing use of AI in SaaS platforms introduces new benefits, like automation and personalization. However, this also leads to higher infrastructure costs due to expensive computing resources like GPUs, high-volume data usage, and the need for highly skilled professionals to build and maintain AI systems (<https://www.paddle.com/blog/saas-market-report-q1-2025>, assessed on 1 June 2025). These trends in the SaaS industry indicate that SaaS providers face increasingly complex decisions. As a result, the role of data-driven decision support systems becomes critical in helping SaaS vendors make informed, timely, and cost-effective choices.

At the core of modern SaaS ecosystems, vast amounts of data are generated, regarding user behavior, subscriptions and system performance [7]. SaaS data can be leveraged for deriving insights and identifying patterns to support decision-making for SaaS providers. For example, machine learning (ML) algorithms can be employed to create churn prediction models for user churn prediction [8]. Given that information, SaaS providers can act with a specific incentive to reengage those users. To this end, the aim of this survey is to review research in data-driven decision support in SaaS applications. In particular, the goal of this work is to identify research that utilizes data to support SaaS vendors towards reducing churn and increasing user engagement, satisfaction and loyalty.

Although related works are already present in the literature [9–11], this study stands out by offering a comprehensive perspective on leveraging SaaS data to support decision-making for SaaS vendors to sustain user loyalty and satisfaction. Unlike prior research that often focuses narrowly on specific tasks such as churn prediction, this work provides a holistic examination of strategies to maintain a stable user base. Specifically, the diverse objectives of existing approaches are analyzed, as the variety of employed data sources, and the comparative effectiveness of different machine learning methods on multiple occasions. Moreover, the ways the outputs of these approaches, ranging from visualizations

to actionable insights, are presented to SaaS vendors to facilitate spot-on decision-making, are also discussed.

The rest of the paper is structured as follows: Section 2 presents related works and highlights the contributions of this paper. Section 3 summarizes the research methodology, while Section 4 presents the results. Discussions and conclusions are provided in Sections 5 and 6, respectively.

2. Related Work

The current work constitutes a scoping review of research to assist SaaS vendors in decision-making based on the data that SaaS businesses generate. Therefore, the aim is to identify research efforts that perform literature reviews on the same subject. To the best of our knowledge, there is no exact type of review found in the literature. However, there are literature reviews that are related to the subject, yet with different focus. For example, the authors in [9] performed a review on machine learning methods for churn prediction covering the years between 2015 and 2023. Although their work provided insights on the use of ML for churn prediction highlighting the trade-offs of different ML methods, the focus was only on churn prediction, and it did not address other use cases such as user segmentation or customer lifetime value prediction which are critical to support business owners on decision-making. Moreover, the review focused on telecommunications, finance and online gaming industries, regardless of the type of businesses and the adoption of cloud-based models and SaaS specifically. To this end, our work constitutes a review tailored to support businesses that operate based on the SaaS cloud model providing a more holistic approach by covering a wide range of use cases, important to retain a strong customer base.

Similarly, the authors in [10] conducted a general review of decision analytics approaches for cloud computing. It's also worth noting that their work was published more than 10 years ago, in 2014, and therefore it does not cover current research efforts. The authors focused on decision support for cloud computing for problems like service selection and pricing, rather than discussing decision support for user retention and engagement. Moreover, there was no description of the systematic review research methodology and no information or statistics regarding the number of papers included in their review. Moreover, the authors considered papers with methods that were not data-driven, like heuristics or ontology-based. The review presented in [11] addressed a broad range of different topics AI, ML, Business Intelligence (BI) and SaaS at a high level without going into detail. Regarding user behavior, authors discussed prediction strategies and personalization of user experience, however, without going into specific insights and recommendations. A systematic review methodology was also missing.

Another paper [5] published in the same year (2014) covered the technical challenges and perspectives for SaaS such as multi-tenancy and scalability. That review also lacked a research methodology and did not mention how the reviewed approaches were selected. Performance comparisons were not included, and only a qualitative analysis was performed. Lastly, there was no focus on user interaction subjects like user retention and churn. Finally, the research presented in [11] addressed AI adoption for BI in SaaS. Although it was mentioned that a literature review was conducted, only 8 papers were included in their analysis.

To this end, compared to the existing bibliography, the current work:

- Presents a systematic review process.
- Includes only data-driven methods.
- Presents of performance comparison to support model selection.

- Discusses of challenges the SaaS vendors face regarding maintaining a core user base like churn, user engagement and user retention.

Moreover, the contributions of this work can be summarized in the following distinct points:

- A comprehensive review of research initiatives aiming to support decision-making for SaaS providers, expanding beyond single-task approaches to address broader strategic goals.
- An overview of various data inputs utilized across studies, identifying frequently used data sources and highlighting cases where multiple kinds of data were integrated for more robust insights.
- A comparative evaluation of employed machine learning techniques, analyzing their relative strengths, weaknesses, and performance in different SaaS decision-making contexts.
- An overview of user-faced output types proposed by researchers to help SaaS providers take effective, data-driven actions.
- A synthesis of best practices and key takeaways from the literature, offering actionable recommendations for enhancing user retention and ensuring the long-term sustainability of SaaS business models.

3. Research Methodology

The research questions that we aim to answer in this survey are the following:

RQ1: What is the main focus of the papers in the area of data-driven methods for supporting decision-making in SaaS?

RQ2: What kind of data is utilized?

RQ3: Are machine learning methods employed and how effective are they?

RQ4: How are the results presented to SaaS providers to support decision-making?

The survey was conducted following the PRISMA-ScR methodology. The research was conducted using the Scopus and Scholar bibliographic databases that contain papers from libraries like Springer, Elsevier, IEEE, etc. The following query was posed on 5 March 2025 to Scopus:

“web usage mining” OR “customer segmentation” OR “log analysis” OR “churn” OR “user engagement” OR “customer lifetime value” OR “user lifetime value” OR “user segmentation”) AND “SaaS” AND PUBYEAR > 2013 AND PUBYEAR < 2026.

This query returned 426 results. Screening was performed by reading the title and abstract in order to identify the completely irrelevant ones. This process resulted in 147 papers. These papers were searched and retrieved in order to read them thoroughly. During this second screening phase, there were excluded all papers that were not relevant to our subject. Relevancy was based on three main rules: papers should be about SaaS, decision-support and based on data. Thus, the excluded papers of the second phase fall into one of the three categories:

- Not related to SaaS;
- Although related to SaaS they were not about supporting decisions;
- Not data-driven, theoretical.

This process resulted in 22 papers that were included in our survey. A supplementary search was conducted in Google Scholar using keywords like “churn prediction SaaS”, “user engagement SaaS” and “user lifetime value SaaS” resulting in 6 more papers. Thus, a final list of 28 papers was formed that was analyzed further. Figure 1 illustrates the research process followed.

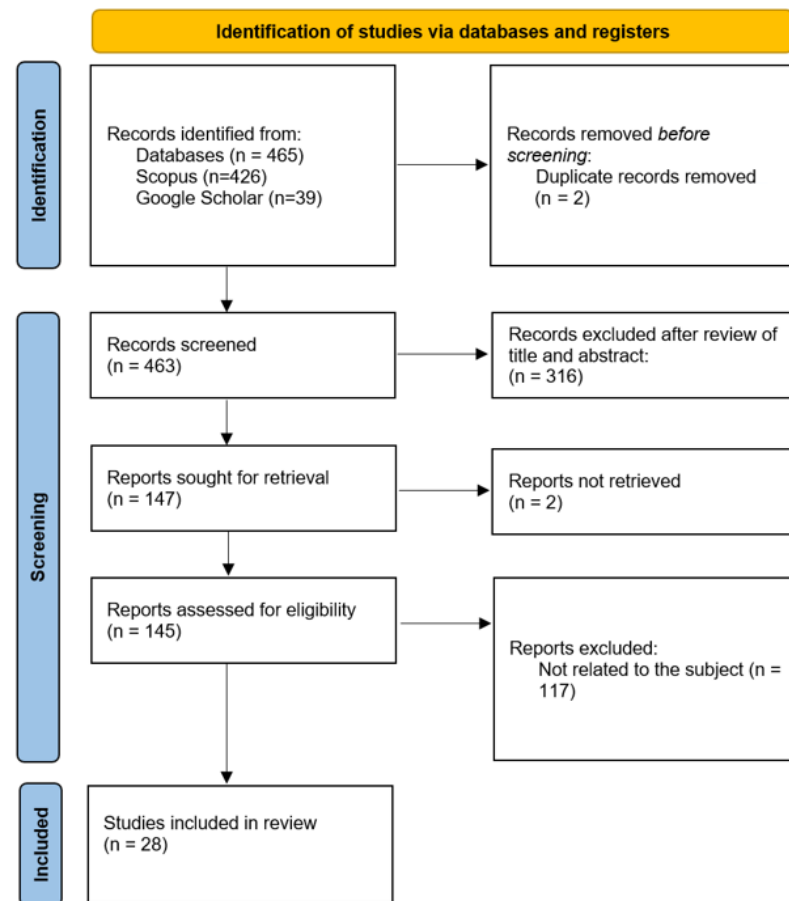


Figure 1. PRISMA flow diagram.

At this stage, it is important to acknowledge the limitations of this research. The research was limited from 2014 to 2025, so as to cover a full range of 10 years, as well as due to the fact that SaaS adoption began to rise mainly after 2014, where the main volume of literature was located. However, the latter did not affect our research findings, since the main volume of the research was located after 2014 and especially from 2020 onwards. Moreover, the database coverage was limited to Scopus and Google Scholar. While other databases also exist, such as Web of Science or PubMed, the use of both Scopus and Google Scholar could improve coverage and reliability, as they both cover a wide range of peer-reviewed sources.

4. Data-Driven Decision Support in SaaS

4.1. Main Focus

The SaaS model is mainly offered by subscription which is the main source for revenue generation for such businesses. Having a strong user base is crucial for the viability of SaaS businesses. Therefore, it is very important for SaaS to maintain a small percentage of users that cancel their subscriptions (churn). For SaaS businesses, even a small reduction in churn can lead to significant profit growth and a healthier and long-term business. For this reason, many researchers focus on predicting churn in SaaS businesses and on identifying the best performing machine learning algorithms for each case.

Figure 2 summarizes the results regarding RQ1. As can be seen in Figure 1, most of the reviewed papers (53.57%) focused on churn prediction. This comes as no surprise since high churn rates cannot sustain a SaaS business model. The churn rate is the number of users that leave the SaaS divided by the total number of users. Churn prediction refers to predicting if a user is about to churn. The authors in [12,13] focused on applying and testing

various algorithms fed with usage data to evaluate their performance in churn prediction, in the context of SaaS, while research presented in [8,14–17] followed the same principle but shifted their focus towards the Business to Business (B2B) SaaS business model. All those studies aimed to point out key features that churning customers possess and offer SaaS providers actionable insights for improving their customer retention strategies.

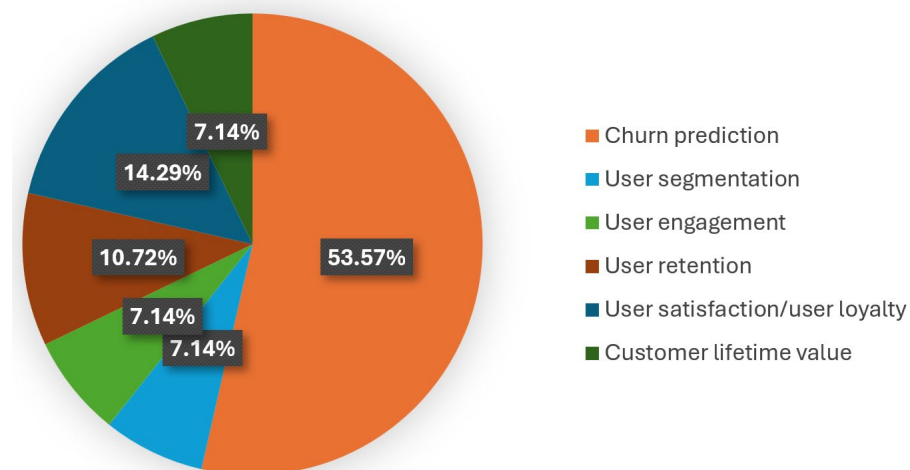


Figure 2. Main focus of the reviewed papers (RQ1).

Further research on this topic included in [18,19] focused on comparing multiple machine learning algorithms to figure out which one was best performing for churn prediction in the SaaS cloud service model. The authors in [18] developed a churn prediction system to predict customer attrition for cloud-based service providers (CSPs), attempting an early identification of customers at risk of canceling subscriptions. Churn prediction in Customer Relationship Management systems (CRM) using hybrid machine learning models was addressed in [19]. Apart from the comparison between machine learning methods, the authors in [20] chose to use machine learning to analyze and monitor customer satisfaction from support tickets and predict whether a client is going to renew their cloud service subscription. Accordingly, the research presented in [21] experimented with the creation and application of a hybrid machine learning framework designed for real-time prediction of user events in SaaS products such as subscription cancellation, user interactions, and task abandonment during user sessions.

Several research efforts dealt with predicting churn and improving retention specifically for online games based on the SaaS model. The strategy employed in [22] aimed to identify churners in earlier stages of a mobile freemium game, even after the tutorial phase, using minimal early-stage user data, featuring a Transformer-based architecture (FT-Transformer) tailored for tabular data. The research presented in [23] focused on predicting player churn and disengagement within a 14-day window in a freemium online strategy game using machine learning. Specifically, it aimed to identify the most effective machine learning techniques and labeling approaches for early detection of players at risk of leaving the game, enabling proactive retention strategies. In [23] the authors distinguished between churn (permanent departure) and disengagement (reduced activity), providing a comprehensive comparison of methods tailored to the gaming industry. The authors of [24] extended their research to a larger time span, focusing on improving churn prediction in casual freemium games by combining sequential (temporal) and aggregated (static) data using different neural network architectures. Their study investigated how integrating these two types of data can enhance prediction accuracy compared to models that rely solely on sequential or aggregated data while also addressing the challenge of predicting

churn in non-contractual contexts (for example in mobile games), where players could leave without explicit signals.

Additionally, the research conducted in [25] focused on enhancing retention analysis in freemium role-playing games (RPGs) by modeling players' motivation, progression, and churn. Particularly, it aimed to understand how in-game behaviors (engagement, collaboration, and achievement) at different levels influence dropout rates, and how these interdependencies could be leveraged to predict player retention more accurately. As player retention in the video game industry is the key to its ongoing prosperity, work in [26] explored how video game operators can retain players in subscription-based games by dynamically adjusting the quality of the game over time. The study developed a dynamic programming model that considered players' memory of past service experiences and network externalities (the phenomenon where the value or utility of a product or service increases as more people use it) to determine the optimal data-driven decisions to maximize long-term profits.

In order to better understand and adapt to clients' needs in SaaS services it is important to categorize them into groups to be able to provide them personalized attention. An iterative mixed-method approach for creating user personas in the context of B2B SaaS products was presented in [27]. That work also aimed to demonstrate how the generated user personas could be practically applied as an indicator to improve the design, development, and prioritization of features in a B2B SaaS product. Another research on customer segmentation was presented in [28], particularly exploring how behavioral customer segmentation and app usage analysis can be leveraged to predict customer interest in premium subscriptions and identify key influencing factors to improve subscription conversion rates for mobile apps. User conversion from free trials to paid subscriptions was the main focus of [29], examining how marketing interactions (ads, messages, emails, etc.) and their type of content along with free-trial usage behaviors could influence users' decisions. Further insights regarding free trials were showcased in [30]. The study highlighted the importance of free trials as a customer acquisition strategy in the SaaS industry and analyzed how customer acquisition, retention and profitability were affected by the duration of free trials to discover their optimal design.

Establishing a long-term relationship with customers requires keeping them engaged and satisfied with the services they are getting. The authors in [31] examined how visual and functional aspects of user interface (UI) design could influence user experience (UX). Advancing further on the subject of customer loyalty, the authors in [32] followed a data-driven approach to discovering the determinants of customer loyalty in the B2B SaaS industry. That study explored how transactional and behavioral data (like platform engagement, and communication frequency) could be leveraged to predict customer loyalty without relying on traditional surveys. The research presented in [33] proposed the design of an adaptive negotiation mechanism to allow providers to interact with clients and dynamically manage service quality, user satisfaction, and resource costs in cloud environments.

A very impactful factor in keeping customers from churning is whether these individuals discover the "aha moment" and acknowledge the return on their investment in SaaS services. The "aha moment" refers to the point when users understand the value of the software product, which is crucial for enabling customer activation and retention. The authors of [34] focused on identifying the "aha moment" of B2B SaaS customers using process mining techniques. Specifically, this study investigated how customers switch from the activation phase to the retention phase in the AARRR (Acquisition, Activation, Retention, Revenue, Referral) model by identifying and examining behavior patterns of clients. In order to provide value to their customers, customer feedback must be taken into consideration, so that providers can offer a more personalized experience and cater to their

clients' needs. The research presented in [35] explored structured approaches for collecting and integrating customer feedback in SaaS companies operating in the B2B market and ways of integrating customer knowledge into software development processes.

SaaS pricing also takes its toll on user satisfaction as it affects SaaS users directly. The study presented in [36], however, showcased a data-driven framework integrating real-time usage tracking, machine learning for demand prediction, and customer-centric billing adjustments to optimize revenue for providers while enhancing user experience and customer loyalty. Additional factors influencing usage continuance in SaaS applications, particularly after the initial adoption phase were demonstrated in [37], in an attempt to improve user retention and reduce churn by enhancing usage penetration (the number of users actively engaging with the software).

Finally, another problem that the research addressed was the Customer Lifetime Value (CLV) prediction. CLV prediction models predict the future revenue that customers may generate. The authors in [38] proposed a novel machine learning framework for predicting CLV in the context of B2B SaaS companies and described several business applications where CLV predictions were used to enhance marketing expenditures, improve Return on Investment (ROI), and provide essential insights for management decision-making in this context.

4.2. Data Sources

In order to compare and evaluate algorithms for predicting churn in SaaS businesses, researchers chose to utilize various types of data sources for their experiments. As per the research question RQ2, the most common category among studies [8,12–19] is usage behavior data. All those studies incorporate some form of user interaction or system activity logs like logins, sessions, file uploads, application usage, number of users and actions performed.

Another common type of data used in [8,12–14,16,17] is transactional and business metrics including monetary values (monthly charges, total expenses, amount spent), subscription status and billing information. Additionally, temporal data, contractual data and customer lifecycle indicators appeared in [12,13,15,17–19] concerning customer tenure, days to expiry, subscription age and monthly snapshots or observation windows while in studies [8,16–19] customer demographics and attributes were also utilized (company size, region, industry, onboarding status or business age).

The authors in [18] also used billing and loyalty program data like loyalty program status and billing cycles. At last, customer support interaction data were used in [15] (support tickets, resolution times) and [16] (support case logs) while customer satisfaction metrics such as Net Promoter Score (NPS) and call quality were specifically included in [17]. Furthermore, regarding churn prediction, the approach followed in [20] used support and subscription metadata while the study [21] focused on clickstream data combined with dynamic user profiles (subscription status, tenure, demographics).

Studies that focused on predicting churn in games featuring a subscription model as in [22–24], shared a strong emphasis on event logs, session data, and gameplay behavior and chose to utilize behavioral data (gameplay actions, tutorial engagement, session metrics, clicks, purchases, logins) to design their churn predicting models.

Focusing on improving retention of freemium online games, the study [25] conducted its research by utilizing player behavioral data (engagement, achievements, collaboration, progress, dropouts), demographic and contextual data (gender of game characters, geographic location, time of play), and some game-specific metrics. Moreover, the data used in [26] was derived from a simulated dataset containing synthetic data (simulated quality levels, utility functions, and cost structures).

Data in [27] are derived from surveys, interviews and web behavioral data (time spent per page, clicks per feature, aggregated over time) and are used in customer segmentation algorithms. On the same page, work [28] used demographic features (age, gender, state), behavioral features (engagement level, time spent, number of screens viewed, clicks, sessions) and transactional features (maximum budget, maximum return on investment, number of purchases) for customer segmentation.

For the purpose of discovering drivers of user loyalty and improving user satisfaction and engagement, studies [29–32,34,37] leveraged mainly behavioral and usage data (login frequency, feature usage, interaction patterns to understand user engagement, product adoption over time). However, in [35,37] interview and survey data were used, providing context on user perceptions, satisfaction, and demographic characteristics. In studies [34,35], system logs and event data were used and specifically in [35] logs are used alongside direct feedback like surveys and support tickets. Marketing, free trial, and transactional data (ad impressions, message types, marketing variables and trial duration variations) are utilized by researchers in [30,33] while in [36] simulated data were employed (transaction volumes, volatility, user-centric metrics, user profiles, negotiation dynamics) to explore hypothetical user behaviors and negotiation dynamics in scalable, controlled environments. Last, studies [32,35] utilized rich, multi-source data integrations, combining CRM, financial, usage, and external firmographic data in order to better model loyalty and customer relationships.

Finally, research [38], which focused on predicting CLV, employed various types of data to achieve its goal. Those data included revenue data (monthly recurring revenue, historical billing data), product license data (product types, acquisition channels, license terms), product usage data (feature adoption, user activity logs, engagement metrics), firmographic data (company size, industry, geography, employee count) and customer segments based on behavior. Table 1 summarizes the types of input data used in the reviewed approaches.

Table 1. Types of data sources used by the reviewed approaches (RQ2).

Ref.	Usage Behavior	Transactional/ Business Metrics	Customer Profile	Financial Data	Customer Support	Satisfaction (e.g., NPS)	Survey/Interview	Marketing
[18]	✓		✓	✓				
[24]	✓							
[12]	✓	✓	✓					
[14]	✓	✓						
[22]	✓							
[19]	✓		✓					
[25]	✓		✓					
[13]	✓	✓	✓					
[20]					✓			
[15]	✓		✓		✓			
[8]	✓	✓	✓					
[23]	✓							
[21]	✓		✓					
[38]	✓	✓	✓	✓				
[29]								✓
[30]	✓						✓	✓
[31]	✓		✓				✓	
[26]			✓			✓		
[33]	✓					✓		
[32]	✓							
[27]	✓						✓	
[37]	✓		✓				✓	

Table 1. Cont.

Ref.	Usage Behavior	Transactional/ Business Metrics	Customer Profile	Financial Data	Customer Support	Satisfaction (e.g., NPS)	Survey/Interview	Marketing
[16]	✓	✓	✓		✓			
[28]	✓	✓						
[35]	✓				✓	✓	✓	
[17]	✓	✓	✓			✓		
[36]		✓		✓				
[34]	✓	✓						✓

4.3. Machine Learning Methods

Regarding RQ3, most of the studies relied solely on machine learning methods to conduct their research and generate their expected results. Among the papers focused on churn prediction, Random Forest (RF) [39–41] was one of the most widely used methods, appearing in studies [8,13–19,23,24]. In the study [18], RF with 64 estimators achieved an excellent performance of 98.8% accuracy, 0.997 Area Under the Curve (AUC), and strong F-measures (0.989 non-churn, 0.981 churn). In studies [8,14], RF also stood out with 87% testing accuracy, while being resistant to noise and overfitting. A study [13] reported exceptionally strong RF results as well, with F1-score of 92.6% and recall of 91.6%, while a study [15] found that RF had the highest precision (0.80) among the models tested, though recall lagged behind XGBoost. On the contrary, RF underperformed in the study [12], with an AUC of ~0.5, and adding PCA (Principal Component Analysis) did not have any impact on the results. In work [23], RF achieved AUC > 0.99 and 97% accuracy, outperforming all other models across various labeling schemes like sliding windows and activity quartiles while in the study [24] RF was used as a baseline combining sequential and aggregated data, where it achieved an AUC of 0.72, which was notably lower than the hybrid Long Short-Term Memory (LSTM) models, indicating its limitations in modeling temporal dynamics in sequential data. In the study [17], RF showed only moderate success with AUC-ROC ~0.65, though its performance improved slightly when paired with SMOTE-Tomek for class imbalance handling. Figure 3 presents the number of papers that successfully employed specific machine learning algorithms. Thus, it contains the number of papers to which the specific machine learning algorithm noted the best results. RF was successfully used in more works than any other machine learning algorithm.

Extreme Gradient Boosting (XGBoost) [42] was tested in studies [12,15,16], consistently delivering top-tier results. In research [12], XGBoost achieved the highest AUC (0.7941) and, after threshold tuning, improved sensitivity to 74%. Study [15] reported ROC AUC of 0.86, recall of 0.85 and F1-score of 0.82, outperforming other models such as Random Forests and Logistic Regression and identified ticket resolution time and license age as key features. Study [16] found XGBoost and Logistic Regression achieved comparable AUC (~60%), but XGBoost was less profitable than logistic regression when measured by Expected Maximum Profit. XGBoost and Gradient Boosting Decision Trees (GBDT), were also used in [22] with XGBoost emerging as the best-performing model with 98.8% AUC, while GBDT followed with 93.4% AUC.

Decision Trees (DT) were used in studies [8,13,14,16,19]. In work [13], DT was the best performer using Chi-squared-selected features, with 88.2% F1-score and 94.4% recall. In contrast, studies [8,14] revealed DT's tendency to overfit, achieving perfect training accuracy but only 76% on test data. DT also served as a baseline in [16,19], with moderate performance compared to ensemble and hybrid models.

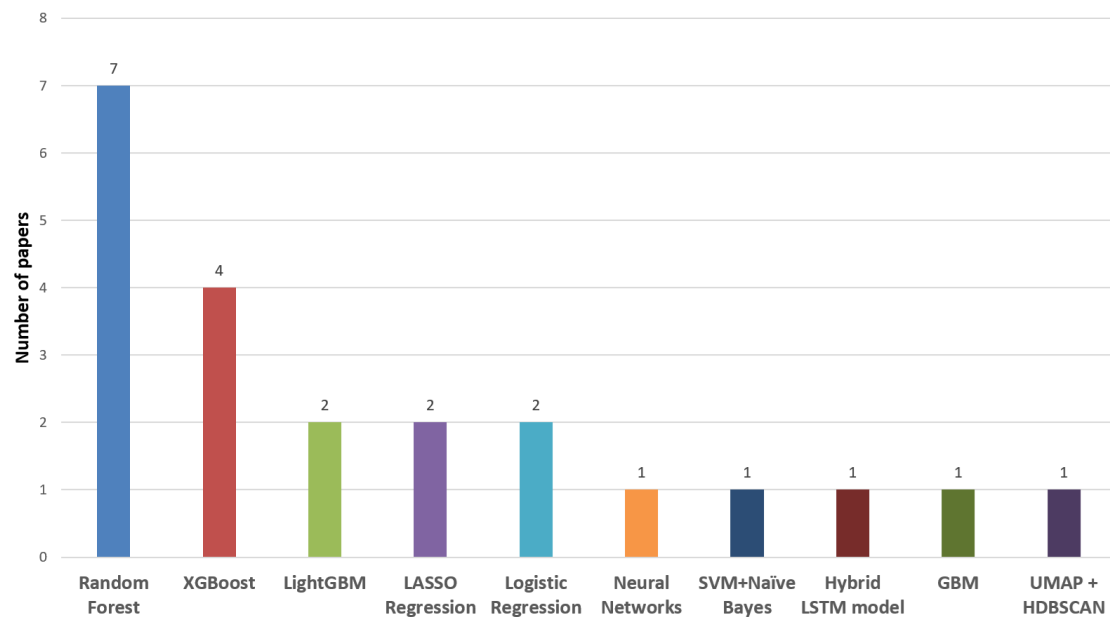


Figure 3. Machine learning methods proposed to use by the reviewed papers.

Logistic Regression (LR) [43] was applied in studies [8,12,13,15–17], mostly as a baseline due to its interpretability. Its AUC ranged from 0.71 to 0.74 in studies [8,12], and it consistently offered stable, if lower, performance. In work [16], LR delivered strong profit-based performance, leading in Expected Maximum Profit and Top Decile Lift when paired with usage-based features. In the research [17], LR initially had 97% accuracy but 0% recall until SMOTE-Tomek was applied, improving recall to 57% (but at the cost of reduced precision).

Support Vector Machines (SVM) featured in studies [8,13,14,16,19]. SVMs often struggled—studies [8,14] reported 100% training accuracy but just 63% test accuracy, indicating severe overfitting. However, the study [19] successfully employed a hybrid model combining SVM with Naïve Bayes, achieving 95.67% accuracy, 94.3% precision, and 95.65% recall, outperforming other classifiers. In work [16], SVM delivered strong profitability metrics, rivaling LR in Expected Maximum Profit.

Neural Networks (NN) (including MLPs and deep learning) appeared in [8,14,15,18,20,23,24]. A study [18] reported 96.5% accuracy for a Multilayer Perceptron (MLP), which, while lower than RF and AdaBoost, still offered valuable comparative insights. In works [8,14], deep neural networks (including TensorFlow models) achieved around 82% test accuracy, with one model identifying subtle churn signals like low session frequency and high pricing. MLP was again tested in [15], performing adequately but trailing ensemble methods like XGBoost and Gradient Boosting Machine (GBM). In [24], a non-sequential NN using only aggregated features yielded lower performance (AUC not specified), underperforming compared to LSTM-based hybrids. In research [20], sentiment modeling with LSTM networks produced learned sentiment features from support ticket data, which when combined with metadata features, led to a noticeable +5% increase in accuracy and the highest recall, critical for predicting subscription renewals. Single-layer neural networks were employed in [23] achieving AUC = 0.987, though their performance degraded over time due to staleness, making them less ideal for evolving user behavior.

AdaBoost [44] and other boosting techniques such as GBM [45] were included in studies [15,18]. AdaBoost in [18] was the second-best performer with 98.4% accuracy and 0.995 AUC, while in [15], boosting methods (especially XGBoost) consistently outperformed simpler classifiers across recall, AUC, and F1-score.

Naive Bayes [46], though rarely a top performer, was evaluated in [8,14,19]. It performed poorly when used alone in [8,14] (accuracy ~69–71%), but in [19], it formed part of the high-performing hybrid model with SVM.

Transformer-based models [47] were explored in the study [22]. The FT-Transformer, built for tabular data with embedded categorical and numerical features, achieved 86.79% accuracy, 88.75% recall, and 94.9% AUC—a strong improvement over older works (AUC ~71–77%), though it was still outperformed by XGBoost (AUC 98.8%, F1 94.81%, accuracy 94.8%) in the same study.

Hybrid Models were explored in [19], where an SVM-Naive Bayes combination led to superior performance across all key metrics, indicating the strength of ensemble learning strategies, especially in reducing false positives/negatives for CRM applications. Additionally, ensemble and hybrid architectures were core in studies [20,21,24]. Hybrid LSTM architectures combining sequential and static aggregated data outperformed single-source models [24]. Study [21] introduced the BBE-LSWCM ensemble, integrating LightGBM, BiLSTM, and LR to achieve 60% better decile lift ($DL1 = 3.197$) and 25% higher AUROC than standalone BiLSTM, while maintaining reasonable latency. A modular fusion of LSTM-learned sentiment features and handcrafted metadata yielded the highest accuracy and recall, demonstrating the value of combining deep learning with traditional features [20].

The researchers in [26] employed a dynamic programming model to optimize player retention over time alongside a Q-learning algorithm [48] to simulate offline decision-making and identify convergence paths for both actual and perceived quality. The results indicated that both game quality and perceived quality stabilized at high levels regardless of initial quality. However, the speed of convergence depended on starting conditions, with lower initial quality ($x_0 = 0.4$) taking 20 periods to converge compared to 14 periods for $x_0 = 0.6$. Interestingly, when players initially overestimated quality, perceived quality followed a “high-low-high” trajectory.

The authors of [27] employed an iterative clustering approach starting with PCA for dimensionality reduction and K-means clustering, later switching to Uniform Manifold Approximation and Projection (UMAP) and Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) for more effective handling of sparse and non-normal data distributions. After four iterations, the final model identified six meaningful personas (e.g., Uploaders, TV Builders), which were validated through interview-based triangulation. Earlier clustering failed due to overlapping behaviors and sparse data, but the refined method succeeded in differentiating user types by combining behavioral data with qualitative insights. At last, in the study [28] a more traditional machine learning approach was adopted for behavioral customer segmentation to predict subscription conversion by evaluating several classifiers, including Decision Tree, K-Nearest Neighbors (KNN), Naive Bayes, RF, LR, SVC, and XGBoost. Among these, XGBoost achieved the highest performance, with an accuracy of 79%, precision of 80%, recall of 76%, and an F1-score of 78%. Cross-validation further confirmed the generalizability of the model with a mean score of 78.5%. However, the model exhibited a slightly higher rate of false negatives than false positives, a relevant trade-off when considering conversion sensitivity.

Studies [30,32,36] utilized machine learning and statistical modeling to personalize user experience and predict outcomes. Study [30] employed Lasso regression [49] to personalize free trial lengths, showing a 6.8% subscription increase compared to a 30-day baseline. Simpler uniform policies like a 7-day trial still yielded a 5.59% gain. Among alternatives, Lasso outperformed random forests and causal forests, which struggled with overfitting and poor personalization. Similarly, a study [36] implemented logistic regression, random forest, and gradient boosting for dynamic pricing in usage-based models. While logistic regression achieved perfect accuracy on simulated data, real-world

applicability favored gradient boosting and random forest, with accuracies around 0.77 and 0.75, respectively. Study [32] leveraged hierarchical logistic regression within the Cross Industry Standard Process for Data Mining (CRISP-DM) framework to predict customer retention and cross-buying. Retention prediction reached 93.7% accuracy (F1: 94.38%), while cross-buying was less predictable (accuracy: 78.9%, F1: 20.60%).

Finally, in the work presented in [38], the authors used a hierarchical ensemble machine learning framework for predicting CLV in B2B SaaS companies. The CLV prediction was framed as a lump sum regression task rather than a traditional time-series forecasting problem, enabling the use of rich supervised learning models like LightGBM, XGBoost, LASSO regression, KNN, and Auto ARIMA. To handle limited and drifting historical data, a two-step hierarchical model was built: first predicting over a short time horizon, then mapping it to a longer horizon. They also adopted an ensemble approach, segmenting customers based on key features (like size) and fitting different models for different groups (LightGBM for smaller customers, LASSO for enterprises). In performance tests, the lump sum regression models outperformed traditional time-series models by 2 to 5 times across metrics like Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and symmetric mean absolute percentage error (SMAPE), with LightGBM giving the best results.

Few researchers, however, opted to utilize several methods that rely on statistical, econometric or heuristic models. In detail, study [34] deployed Process Mining [50], specifically Heuristic Miner and Fuzzy Miner, to track user behaviors leading to “Aha! moments”. Through data cleaning and retention clustering, it identified key activation actions by reducing model complexity by 62%. However, manual data processing was time-intensive.

Linear Mixed Models (LMMs) were utilized in [37] to evaluate usage continuance. It accounted for repeated measures and individual client variability through fixed and random effects. For existing clients, the model achieved an RMSE of 5.23, though it struggled with new clients (RMSE: 13.93), indicating a gap in generalization. Activation measures (banners) significantly increased usage (+5.2%).

PLS-SEM (Partial Least Squares Structural Equation Modeling) was employed in [31] to validate the effect of UI design on satisfaction and loyalty. The results showed strong model reliability (Cronbach’s alpha: 0.837–0.860) and high explanatory power ($R^2 \sim 0.50$). All hypotheses (UI design $\rightarrow$ Satisfaction $\rightarrow$ Loyalty) were statistically supported.

Focusing on the topic of customer segmentation, the study [25] introduced a joint modeling framework that combined Generalized Linear Mixed Models (GLMMs) [51] to track players’ in-game behavioral patterns with a Shared Parameter Model (SPM) [52] that linked these behavioral groups to the probability of dropout. This integrated approach significantly outperformed traditional models like Cox regression, improving retention prediction by 63.2% at higher player levels and reducing RMSE for lifetime engagement by 25.5%.

Authors in [29] took a more econometric route, applying a Dynamic Probit Model [53] with copula-based corrections for endogeneity. It incorporated exponentially decaying goodwill stocks to model user touchpoints and usage effects. In this study regularization via Lasso/Ridge regression was also utilized, addressing multicollinearity and improving model stability. The study achieved a 9.3% error reduction in predictions and key findings emphasized that consumer-initiated touchpoints and persuasive content boosted conversions to premium users, while firm-initiated ads had negative effects.

Study [33] introduced the AQUAMan negotiation mechanism, combining quantile estimation, opponent modeling, and surplus redistribution to manage service acceptability. Under both normal and extreme loads (up to 2500 users/min), it maintained high acceptability rates (95%+ and 93.3%, respectively) while keeping costs and overhead manageable.

At last, study [35] outlined a range of qualitative (interviews, questionnaires, incident reports) and quantitative (A/B testing, in-product surveys and online ads, crowdfunding and crowdsourcing platforms, click-based user data collection) methods deployed for collecting and integrating customer feedback, as well as insights into their performance across several companies. This study concluded that there's no single perfect approach and that the choice and success of methods depend heavily on the company's size, the structure of the product offering (custom vs. SaaS), the type of customers (B2B vs. B2C), the stage of development and the internal communication and data integration structure.

Table 2 summarizes the results regarding the use of machine learning in the reviewed approaches for SaaS.

Specifically, Table 2 includes the proposed ML algorithm for the reviewed papers along with the compared methods. Results are split into two categories, indicating the most commonly used metrics and miscellaneous across the reviewed papers. Additionally, information about the used datasets and employed validation methods is included.

Table 2. Machine learning methods used for decision support in SaaS (RQ3).

Ref.	Proposed Method	Compared Methods	Evaluation Results		Validation Method	Dataset
			Common Metrics	Miscellaneous Metrics		
[18]	Random Forest	Neural Networks, AdaBoost	Random Forest: 0.997 AUC, 0.988 Accuracy, 0.989 F-measure for Non-Churn Class and 0.981 for Churn Neural Networks: 0.968 AUC, 0.965 Accuracy, 0.975 F-measure for Non-Churn Class and 0.946 for Churn AdaBoost: 0.995 AUC, 0.984 Accuracy, 0.989 F-measure for Non-Churn Class and 0.974 for Churn	-	Training (64%), validation (16%), and test (20%) sets	Dataset from a partner company associated with the University of Évora, containing 196,977 instances corresponding to 26,418 service subscriptions
[24]	Hybrid Model (LSTM Hidden State)	Random Forest, LSTM	Hybrid Models: 0.8741 AUC, 0.6953 F1-score, 0.8023 Accuracy LSTM: 0.8592 AUC, 0.6795 F1-score, 0.7900 Accuracy Random Forest: 0.8405 AUC, 0.6414 F1-score, 0.7749 Accuracy	-	10-fold cross-validation	Dataset from player logs of a freemium mobile game developed by Tactile Games including 2,284,238 records of 814,822 unique players
[12]	XGBoost	Logistic Regression, Random Forests	XGBoost: 0.7526 AUC Random Forest: ~0.5 AUC Logistic Regression: 0.7257 AUC	-	10-fold cross-validation	Dataset from a client SaaS company including 76,668 observations of 20 predictor variables

Table 2. Cont.

Ref.	Proposed Method	Compared Methods	Evaluation Results		Validation Method	Dataset
			Common Metrics	Miscellaneous Metrics		
[14]	Random Forest	Decision Trees, Support Vector Machine, Neural Networks, Naïve Bayes, Logistic Regression	Random forest: 0.88 Training Accuracy, 0.87 Test Accuracy Decision Trees: 1.00 (overfitting) Training Accuracy, 0.76 Test Accuracy Support Vector Machine: 1.00 (overfitting) Training Accuracy, 0.63 Test Accuracy Neural Networks: 0.85 Training Accuracy, 0.82 Test Accuracy Naïve Bayes: 0.71 Training Accuracy, 0.69 Test Accuracy Logistic Regression: 0.73 Training Accuracy, 0.71 Test Accuracy	-	train-test split (percentage not mentioned)	A B2B SaaS subscriptions dataset (source not mentioned) including 7044 examples of B2B SaaS subscriptions and 21 variables
[22]	XGBoost	Transformer-based models (FT-transformer), GBDT	XGBoost: 0.948 Accuracy, 0.9545 Precision, 0.9418 Recall, 0.9481 F1-Score, 0.988 AUC Transformer-based models: 0.8679 Accuracy, 0.8477 Precision, 0.8875 Recall, 0.8671 F1-Score, 0.949 AUC GBDT: 0.8561 Accuracy, 0.8755 Precision, 0.8328 Recall, 0.8536 F1-Score, 0.934 AUC	-	train-test split (percentage not mentioned)	Dataset from a mobile freemium game includes data from over 268,370 users
[19]	Hybrid Model (SVM + Naïve Bayes)	KNN, Random Forest, ANN, Decision tree	Hybrid Model: 0.9567 Accuracy, 0.943 Precision, 0.9565 Recall, 0.943 F1-Score ANN: 0.789 Accuracy, 0.8403 Precision, 0.8824 Recall, 0.8608 F1-Score Random Forest: 0.8 Accuracy, 0.79 Precision, 0.80 Recall, 0.79 F1-Score KNN: 0.839 Accuracy, 0.826 Precision, 0.829 Recall, 0.781 F1-Score Decision Tree: 0.9097 Accuracy, 0.9242 Precision, 0.9242 Recall, 0.9242 F1-Score	-	train-test split (80% train—20% test)	Dataset from Kaggle containing subscription details on 7044 customers of a fictional SaaS company

Table 2. Cont.

Ref.	Proposed Method	Compared Methods	Evaluation Results		Validation Method	Dataset
			Common Metrics	Miscellaneous Metrics		
[13]	Random Forest	Decision Tree, Logistic Regression, Support Vector Machine	Random Forest: 0.916 Recall, 0.926 F1-Score, 0.92 Accuracy, 0.939 Precision	-	train-test split (80% train—20% test), 10-fold cross validation	Dataset extracted from the case-study company's database system containing 1788 observations of churn and non-churn samples
			Decision Tree: 0.945 Recall, 0.871 F1-Score, 0.845 Accuracy, 0.809 Precision Logistic Regression: 0.868 Recall, 0.902 F1-Score, 0.896 Accuracy, 0.939 Precision Support Vector Machine: 0.881 Recall, 0.839 F1-Score, 0.814 Accuracy, 0.803 Precision			
[11]	Neural Networks	-	Neural Networks: 0.9694 Accuracy, 0.9651 Precision, 0.9540 Recall	-	Not provided	Dataset collected from a cloud service provider including approximately 700 unique cloud service offering-customer pairs and around 90,000 associated support tickets
[15]	XGBoost	Random Forest, Logistic Regression, Neural Networks, AdaBoost, Gradient Boosting Machine	XGBoost: 0.7956 Accuracy, 0.7916 Precision, 0.8507 Recall, 0.8201 F1-Score, 0.86 ROC AUC Random Forest: 0.7877 Accuracy, 0.8042 Precision, 0.8096 Recall, 0.8069 F1-Score, 0.85 ROC AUC Logistic Regression: 0.7757 Accuracy, 0.7986 Precision, 0.7895 Recall, 0.7940 F1-Score, 0.84 ROC AUC Neural Networks: 0.7835 Accuracy, 0.7910 Precision, 0.8220 Recall, 0.8062 F1-Score, 0.84 ROC AUC AdaBoost: 0.7867 Accuracy, 0.7970 Precision, 0.8191 Recall, 0.8079 F1-Score, 0.86 ROC AUC	-	train-test split (80% train—20% test)	Two datasets provided by a Portuguese software house with the final dataset included 9539 observations from the two datasets combined

Table 2. Cont.

Ref.	Proposed Method	Compared Methods	Evaluation Results		Validation Method	Dataset
			Common Metrics	Miscellaneous Metrics		
[8]	Random Forest	Neural Networks, Decision Trees, Logistic Regression, Support Vector Machine, Naïve Bayes	Gradient Boosting Machine: 0.7935 Accuracy, 0.7946 Precision, 0.8402 Recall, 0.8167 F1-Score, 0.86 ROC AUC	-	train-test split (percentage not mentioned)	User activity datasets (source and number of data not mentioned)
			Random Forest: 0.88 Training Accuracy, 0.87 Test Accuracy Neural Networks: 0.85 Training Accuracy, 0.82 Test Accuracy Decision Tree: 1.00 (overfitting) Training Accuracy, 0.76 Test Accuracy Logistic Regression: 0.73 Training Accuracy, 0.71 Test Accuracy Support Vector Machine: 1.00 (overfitting) Training Accuracy, 0.63 Test Accuracy Naive Bayes: 0.71 Training Accuracy, 0.69 Test Accuracy			
[23]	Random Forest	Neural Networks, Decision Trees, Logistic Regression, Support Vector Machine, Naïve Bayes, Gradient Boosting, KNN	Random Forest: 0.997 AUC Decision tree: 0.987 AUC Neural Networks: 0.994 AUC Gradient Boosting: 0.984 AUC Logistic Regression: 0.967 AUC Support Vector Machine: 0.990 AUC KNN: 0.840 AUC Naïve Bayes 0.887 AUC	-	train-test-validation split (percentage not mentioned), cross validation (folds not mentioned)	Dataset from “The Settlers Online (TSO)”, a freemium online strategy game, including 7439 users and 113,643 observed events.
[21]	LightGBM	Hybrid Models (Neural Network with BiLSTM layers)	LightGBM: 0.690 AUROC, ~30 min Training Time Hybrid Models: 0.591 AUROC, ~2 h Training Time	-	train-test split (500,000 records–200,000 records)	Dataset from QuickBooks Online (QBO) users, including 700,000 combinations of users and reference timestamps.
[38]	LightGBM	XGBoost, Gradient boosting, LASSO Regression, K-nearest-neighbors, AUTO-Arima	-	Performance indexed to LightGBM = 1.0× LightGBM: 1 SMAPE, 1 RMSE, 1 MAE XGBoost: ~1.10 SMAPE, ~1 RMSE, ~1.1 MAE	Not provided	Dataset collected from a well-known B2B SaaS company (number of data not mentioned)

Table 2. Cont.

Ref.	Proposed Method	Compared Methods	Evaluation Results		Validation Method	Dataset
			Common Metrics	Miscellaneous Metrics		
				Gradient Boosting: ~1.2 SMAPE, ~1.1 RMSE, ~1.2 MAE KNN: ~1.25 SMAPE, ~1.2 RMSE, ~1.25 MAE LASSO Regression: ~1.25 SMAPE, ~1.1 RMSE, ~1.2 MAE AUTO-Arima: ~1.75 SMAPE, ~5 RMSE, ~2.5 MAE		
[29]	LASSO Regression, Dynamic probit model with copula corrections	-	-	LASSO Regression reduced MSE to 0.122 Dynamic probit model with copula corrections: −3841 Log-Marginal Density, 7808.5 Deviance Information Criterion	Not provided	Dataset from a U.S.-based multinational computer software company operating on a Software-as-a-Service business model including a sample of 14,989 unique consumers
[30]	LASSO Regression	Random Forest, causal forest, XGBoost	-	LASSO Regression: +6.8% subscriptions XGBoost: +6.17% subscriptions Random Forests: Poor (overfitted training data) Causal Forests: Poor (minimal personalization)	train-test split (70% train—30% test)	Dataset from a fully randomized experiment involving 337,724 unconnected users globally

Table 2. Cont.

Ref.	Proposed Method	Compared Methods	Evaluation Results		Validation Method	Dataset
			Common Metrics	Miscellaneous Metrics		
[16]	Logistic Regression	Random Forest, XGBoost, Decision Trees, Support Vector Machine	Logistic Regression: 0.604 AUC Support Vector Machine: 0.603 AUC Random Forest: 0.594 AUC XGBoost: 0.599 AUC Decision Tree: 0.523 AUC	Logistic Regression: 1.682 TDL, 21,209 EMPB (EUR) Support Vector Machine: 1.590 TDL, 22,566 EMPB (EUR) Random Forest: 11.482 TDL, 15,106 EMPB (EUR) XGBoost: 1.360 TDL, 14,351 EMPB (EUR) Decision Tree: 0.856 TDL, 5809 EMPB (EUR)	cross-validation (folds not mentioned)	Dataset from a European software service provider including 3959 subscriptions
[28]	XGBoost	Random Forest, Decision Trees, Logistic Regression, Support Vector Machine, K-nearest-neighbors, Naïve Bayes	XGBoost: 0.79 Accuracy, 0.8 Precision, 0.76 Recall, 0.78 F1-Score All the other methods where outperformed but their specific results were not provided	-	train-test split (80% train—20% test)	Dataset from Kaggle (fineTech_appData) including 50,000 rows of user information
[17]	Random Forest	Logistic Regression	Random Forest: 0.09 Precision, 0.11 Recall, 0.10 F1-Score Logistic Regression: 0.05 Precision, 0.57 Recall, 0.19 F1-Score The proposed model was better at explaining churn drivers (feature importance) than precise prediction.	-	cross validation (folds not mentioned)	Dataset from Aircall, a Software as a Service company including data from about 5000 customers
[36]	Random Forest, Gradient Boosting Machine	Logistic Regression	Logistic Regression: 1.00 Accuracy, 1.00 Precision, 1.00 Recall (due to dataset simplicity) Random Forest: 0.75 Accuracy, 0.714 Precision, 0.789 Recall Gradient Boosting Machine: 0.77 Accuracy, 0.753 Precision, 0.768 Recall	-	train-test split (80% train—20% test)	A simulation dataset replicating real-world Software as a Service (SaaS) usage patterns (number of data not mentioned)

Table 2. Cont.

Ref.	Proposed Method	Compared Methods	Evaluation Results		Validation Method	Dataset
			Common Metrics	Miscellaneous Metrics		
[32]	Logistic Regression	-	Logistic Regression: 0.9372 Accuracy, 0.9549 Precision, 0.9330 Recall, 0.9438 F1-Score, 0.999 AUC	-	Not provided	Datasets collected from four databases at Digidata, a SaaS company where the project was carried out (number of data not mentioned)

4.4. Form of Outputs Presented to SaaS Providers

Concerning RQ4, researchers chose a wide variety of methods to present their results to SaaS providers in order to support decision-making. As listed below, the categories of those methods include:

1. Visualizations

A significant number of papers utilized visual representations like confusion matrices, Receiver Operating Characteristic (ROC) curves, feature importance plots, retention curves, sentiment plots, time-series graphs and process maps. These visuals help non-technical stakeholders quickly grasp patterns, model strengths, and key insights. They also support internal presentations and stakeholder alignment around key metrics and strategies. Table 3 presents the visualization techniques deployed by the reviewed approaches.

Table 3. Visualizations for decision-making (RQ4).

Ref.	Visualizations
[14]	Accuracy curves to show the impact of tree counts in random forests
[22]	AUC-ROC curves and confusion matrices to validate robustness
[25]	Hazard rate curves and predictive intervals for retention statistics
[13]	Feature importance (e.g., prevPeriodTrans) via bar charts
[20]	Temporal sentiment plots track satisfaction trajectories
[23]	ROC curves and feature plots that highlight “missed days” as top predictors
[21]	(BBE-LSWCM) uses decile lift charts to show a 30% churn reduction in A/B tests
[31]	Path analysis diagrams to map UI design to loyalty
[33]	Time-series graphs to compare adaptive vs. non-adaptive negotiation modes
[34]	Process maps that reveal key activation moment
[37]	Trajectory plots that show activation impacts
[16]	Coefficient plots to illustrate usage data’s impact on churn
[35]	Dashboards to automate feedback summaries and issue prioritization
[36]	Comparative plots that guide model selection via accuracy/recall metrics

2. Simulation/What-If Analysis

Simulation tools model hypothetical scenarios (e.g., pricing changes, trial lengths), enabling SaaS providers to test retention strategies and forecast revenue impact before implementation, supporting data-driven decision-making. Study [18] enabled “what-if” scenarios to test the impact of retention strategies like loyalty extensions, helping providers optimize interventions while the work presented in [25] simulated promotional campaigns, showing a 20% engagement boost from aggressive incentives, and models premium purchase values for revenue forecasting. Additionally, work [29] tested timing strategies, revealing peak conversion windows (e.g., post-trial expiration) and warning

against overusing persuasive messaging and study [30] compared trial lengths (7-day vs. 30-day trial) through simulations, finding shorter trials optimal for beginners but longer trials better for experts. At last, in [26], the authors modeled player retention under quality adjustments, showing that initial quality investments reduced attrition and study [36] simulated dynamic pricing strategies, such as outcome-based billing, to align costs with customer value.

3. Segmentation and Persona Modeling

Studies [23,28,32,38] used clustering and behavioral analysis to segment users or create personas. These help SaaS providers tailor features, marketing, and support strategies to distinct user groups for better engagement. Table 4 includes the segmentation groups used in those studies. Authors in [25] categorized users based on demographics and engagement levels, in the study [23] the authors chose to categorize their users by their activity patterns while [38] chose to group users by CLV. Additionally, the work in [27] created B2B user personas while in [28] the authors created behavioral clusters. Finally, the study [32] segmented users by relationship length and cross-selling dependency.

Table 4. Segmentation and persona modeling (RQ4).

Ref.	Segmentation Groups	Description
[25]	Demographics (gender, geography) and engagement levels	Groups people according to demographics and engagement levels
[23]	Activity patterns (e.g., “economy overview usage”)	Classifies players by activity patterns, using loyalty markers to target interventions
[38]	Customer Lifetime Value (CLV)	Segments customers by CLV, focusing on prioritizing enterprise clients for retention
[27]	B2B user personas (e.g., “Data Sellers”)	Creates user personas with pain points and workflow metrics to guide product development
[28]	Behavioral clusters (e.g., education-focused users)	Identifies behavioral clusters, such as education-focused users, for targeted marketing
[32]	Relationship length and cross-selling dependency	Segments by relationship length

4. Business Impact Metrics

Metrics like CLV, MRR and ROI can be used to quantify churn effects. This enables providers to understand the financial implications of retention efforts and prioritize high-value customer segments. For instance, study [18] reported cost savings per retained customer and revenue protection through churn reduction, while the work in [14] demonstrated ROI improvements via dynamic pricing strategies. Early churn detection and associated acquisition cost reductions were highlighted in study [19] and study [25] showcased a revenue boost from collaboration-driven monetization models. Furthermore, studies such as [8,21] focused on reducing negative MRR churn and increasing intervention acceptance through A/B testing, respectively. The prediction of CLV and the application of marginal ROI formulas were addressed in the work [38], whereas the study [30] illustrated how optimized free trial strategies could enhance both retention and revenue. In addition, study [26] linked quality investment costs to gains in network-driven retention, and study [16] evaluated predictive accuracy in relation to carbon emissions. Studies like [36] connected usage-based billing models to a marked increase in customer satisfaction, while [32] quantified retention differences across countries, providing further insights into localization strategies. Table 5 summarizes the business metrics employed by the reviewed approaches.

Table 5. Business impact metrics (RQ4).

Ref.	Business Metrics
[18]	Cost savings per retained customer, revenue protection from churn reduction
[14]	Dynamic pricing ROI
[19]	Churn rates from early detection, acquisition cost reduction
[25]	Revenue boost from collaboration-based monetization
[8]	Negative MRR churn and CLV
[21]	Churn reduction and intervention acceptance in A/B tests.
[38]	Marginal ROI formulas
[30]	Improved retention and revenue
[26]	Quality investment costs against network-driven retention gains.
[16]	Predictive accuracy against carbon emissions
[36]	Customer satisfaction
[32]	Country-specific retention differences

5. Model Deployment and Integration

Model deployment integrates predictive models into SaaS platforms via APIs or cloud pipelines, enabling automated churn prediction and alignment with CRM features. Building on this, CRM and marketing tools use these insights to personalize campaigns based on churn risk or user behavior. Additionally, toolkits and interactive systems support scenario testing and onboarding analysis, helping teams refine strategies and make data-driven decisions.

For instance, the work in [30] integrated RESTful APIs and the CBAR platform to deliver JSON-formatted real-time risk scores, enhancing both the speed and accuracy of predictions. Centralized Bank Account Register (CBAR) not only ranked subscriptions by risk and value but also enabled targeted actions based on live insights. Similarly, study [12] proposed a standalone app that supported dynamic thresholding and feedback loops, allowing continuous model refinement, and introduced a tool offering live churn predictions for real-time decision-making. This study also highlighted declining login activity as an early warning sign and recommends dynamic threshold adjustment based on marketing capacity.

Several other studies proposed additional enhancements. As shown in [21] an AWS SageMaker pipeline was utilized to support both batch and real-time featurization, enabling flexible data processing, and triggered chat pop-ups for high-risk users to foster immediate engagement. Study [16] recommended leveraging cloud storage and algorithmic optimizations to reduce environmental impact, while work [36] provided pseudocode for integrating real-time data streams with billing systems.

Moreover, the work in [24] aligned predictive models with targeted re-engagement strategies for at-risk users, while [19] fine-tuned retention campaigns by balancing precision and recall, and designed hybrid models to ensure scalability for cloud deployment. Study [29] optimized conversion rates through strategic ad placements and precise timing of messaging. Study [17] ensured that key feature usage data is synced with CRM systems and flags low-usage accounts for proactive customer success outreach.

Dashboards and interactive systems also play a critical supporting role. Study [15] presented a dashboard that visualizes churn risk metrics and integrates real-time alerts directly into CRM dashboards, ensuring that teams can act swiftly on critical issues. Study [15] offered a process mining toolkit that uncovers “Aha!” moments during onboarding, enhancing user activation efforts. Finally, authors in [35] proposed an automated feedback system that prioritizes critical user issues and provides dashboards that visualize feedback trends, helping teams monitor user sentiment and behavior patterns over time.

5. Discussion

In response to RQ1, the analysis revealed that churn prediction is the most frequently pursued objective in decision support systems in SaaS settings. Studies consistently emphasized the importance of identifying early at-risk users in order to interfere with data-driven strategies, therefore improving subscription continuance and reducing revenue loss. Beyond churn, researchers also contributed significantly through user segmentation, which enables the grouping of customers into meaningful behavioral or demographic clusters. This supports personalized product development, targeted marketing, and more responsive customer service. Other studies focused on improving user engagement and predicting CLV, which are key indicators for long-term business sustainability. Additionally, strategic decisions around pricing optimization, free trial policies, and onboarding practices were offered so that SaaS providers are aided in aligning service design with actual user behavior and expectations.

Regarding RQ2, this survey revealed that for the effectiveness of decision-making in SaaS environments, diverse and rich datasets must be used. In most studies, behavioral data, such as clickstream logs, feature usage, and session frequency, emerged as foundational. These were often combined with transactional and subscription-related information, including billing history and account tenure, to provide a fuller picture of user engagement. Studies also made use of demographic and firmographic characteristics, allowing for segmentation by industry, company size, or geographic location. In some cases, sentiment data from support tickets or survey responses were incorporated to capture users' subjective experiences. A few studies employed synthetic or simulated data to test hypothetical scenarios and optimize intervention strategies before deployment. Altogether, the results underscore the importance of integrating both quantitative system usage metrics and qualitative feedback for comprehensive decision support.

Addressing RQ3, it was found that machine learning techniques play a central role in nearly all the examined implementations. RF and XGBoost were the most commonly used algorithms, and they consistently delivered strong performance across various evaluation metrics such as AUC, precision, recall, and F1-score. These models were particularly effective in handling high-dimensional, tabular datasets and offered a balance of accuracy and interpretability. In more complex scenarios requiring temporal modeling or behavioral pattern recognition, hybrid and ensemble models outperformed traditional classifiers. For example, combinations of LSTM networks with LightGBM or structured metadata inputs captured sequential user behavior more effectively than static models alone. While simpler models like LR and DT provided useful baselines, their performance was often eclipsed by ensemble or deep learning approaches. Notably, some studies considered the environmental impact and scalability of different algorithms, highlighting the relevance of computational efficiency and carbon footprint in model selection. Overall, the choice of method was shown to depend on specific task requirements, data characteristics, and operational constraints such as latency or model transparency.

Finally, in response to RQ4, a wide variety of output types were presented to inform and guide SaaS providers. The importance of visual analytics was highlighted, including ROC curves, feature importance rankings, and user behavior maps, which helped non-technical stakeholders interpret model outcomes. Business-oriented metrics such as CLV, MRR, and ROI linked predictions to financial outcomes while simulation tools allowed providers to experiment with trial durations or pricing adjustments before actual implementation, enhancing the strategic value of the DSSs. Furthermore, user segmentation and persona modeling enabled more detailed targeting of user engagement and marketing strategies, while process mining techniques uncovered critical activation patterns such as the discovery of an "aha moment".

Some studies also focused on model deployment strategies, emphasizing cloud integration and scalability. By bridging predictive analytics with tangible business insights and strategic tools, DSS outputs were shown to have a direct impact on operational efficiency, customer satisfaction, and long-term profitability. Moreover, deploying such models as churn prediction ones through the cloud offers the possibility of providing Machine Learning as Service (MLaaS) models specialized for supporting decision-making in SaaS. On the same page, research [54] presented an MLaaS solution for marketing that contained a churn prediction model tested in retail and e-commerce settings.

To synthesize the findings discussed in this review and provide a clearer understanding of the research landscape, Table 6 presents the main insights and findings corresponding to each research question (RQ1–RQ4) addressed in this study. Table 6 offers a compact overview of the focus areas, types of data used, applied machine learning methods, and the nature of decision support outputs in SaaS environments.

Table 6. Cumulative table, findings of RQ1–RQ4.

Ref.	Focus	Data	Proposed Method	Output
[18]	Churn prediction	Usage behavior, customer profile, financial	Random Forest	Simulation/What-If Analysis, Business Impact Metrics, Model Deployment and Integration
[24]	Churn prediction	Usage behavior	Hybrid Model (LSTM Hidden State)	Model Deployment and Integration
[12]	Churn prediction	Usage behavior, customer profile, transactional/business metrics	XGBoost	Model Deployment and Integration
[14]	Churn prediction	Usage behavior, transactional/business metrics	Random Forest	Visualizations, Business Impact Metrics
[22]	Churn prediction	Usage behavior	XGBoost	Visualizations
[19]	Churn prediction	Usage behavior, customer profile	Hybrid Model (SVM + Naïve Bayes)	Business Impact Metrics, Model Deployment and Integration
[25]	Churn prediction	Usage behavior, customer profile	Cox Regression	Visualizations, Simulation/What-If Analysis, Segmentation and Persona Modeling, Business Impact Metrics
[13]	Churn prediction	Usage behavior, customer profile, transactional/business metrics	Decision Trees	Visualizations
[20]	Churn prediction	Customer support	Neural Networks	Visualizations
[15]	Churn prediction	Usage behavior, customer profile, customer support	XGBoost	Model Deployment and Integration
[8]	Churn prediction	Usage behavior, transactional/business metrics, customer profile	Neural Networks	Business Impact Metrics

Table 6. Cont.

Ref.	Focus	Data	Proposed Method	Output
[23]	Churn prediction	Usage behavior	Random Forest	Visualizations, Segmentation and Persona Modeling
[21]	Churn prediction	Usage Behavior, customer profile	LightGBM	Visualizations, Business Impact Metrics, Model Deployment and Integration
[16]	Churn prediction	Usage behavior, transactional/business metrics, customer profile, customer support	Logistic Regression	Visualizations, Business Impact Metrics, Model Deployment and Integration
[17]	Churn prediction	Usage behavior, transactional/business metrics, customer profile, satisfaction	Random Forest	Model Deployment and Integration
[38]	Customer lifetime value	Usage behavior, transactional/business metrics, customer profile, financial	LightGBM	Segmentation and Persona Modeling, Business Impact Metrics
[36]	Customer lifetime value	Transactional/business metrics, financial	Random Forest	Visualizations, Simulation/What-If Analysis, Business Impact Metrics, Model Deployment and Integration
[29]	User engagement	marketing/trials	LASSO Regression	Simulation/What-If Analysis, Model Deployment and Integration
[34]	User engagement	Usage behavior	Heuristic and Fuzzy Mining	Visualizations, Model Deployment and Integration
[30]	User retention	Usage behavior, survey/interview, marketing	LASSO Regression	Simulation/What-If Analysis, Business Impact Metrics
[26]	User retention	Customer profile, satisfaction	Reinforcement learning	Simulation/What-If Analysis, Business Impact Metrics
[37]	User retention	Usage behavior, customer profile, survey/interview	Linear Mixed Models	Visualizations
[31]	User satisfaction/user loyalty	Usage behavior, customer profile, survey/interview	PLS-SEM (Partial Least Squares Structural Equation Modeling)	Visualizations
[33]	User satisfaction/user loyalty	Usage behavior, satisfaction	AQUAman negotiation mechanism	Visualizations

Table 6. *Cont.*

Ref.	Focus	Data	Proposed Method	Output
[35]	User satisfaction/user loyalty	Usage behavior, customer support, satisfaction, survey/interview	Comparative analysis	Visualizations, Model Deployment and Integration
[32]	User satisfaction/user loyalty	Usage behavior, transactional/business metrics, marketing	Logistic Regression	Segmentation and Persona Modeling, Business Impact Metrics
[27]	User segmentation	Usage behavior, survey/interview	UMAP and HDBSCAN	Segmentation and Persona Modeling
[28]	User segmentation	Usage behavior, transactional/business metrics	XGBoost	Segmentation and Persona Modeling

5.1. Insights and Strategic Recommendations

From many of the reviewed systems derived various actionable predictions and recommendations, such as identifying customers at high risk of churning, estimating potential revenue loss, or advising on optimal pricing strategies. One of the most consistent themes across research was the use of machine learning to flag customers likely to churn. Studies such as [14] concluded that newer B2B customers are particularly vulnerable to churn, while those with low monthly payments often exhibit higher loyalty, suggesting that random forests are well-suited for B2B SaaS due to their interpretability and scalability.

Pricing strategies also play a pivotal role in user retention. For instance, research [36] concluded that a significant percentage of users prefer transparent billing and proposed a dynamic pricing formula that factors in base rates, usage metrics, and risk adjustments. The study recommended gradient boosting for complex pricing environments where traditional models may fall short. Additionally, research [32] revealed that multi-product subscriptions improved retention but cautioned against over-reliance on bundling, noting regional variations in loyalty. Further supporting pricing adjustments, research [8] identified that high fees combined with low session activity strongly correlate with churn, suggesting that SaaS providers should consider flexible pricing for low-engagement users.

Additionally, other researchers proposed best-practice policies, like trial durations, onboarding flows, and pricing tactics, to help SaaS providers align product decisions with user behavior and retention trends. Study [30] recommended defaulting to 7-day trials unless segmentation suggests longer ones, while authors of [31] highlighted that intuitive UI design can boost satisfaction and increase loyalty by 41%. Additionally, work [26] advised gradual quality adjustments to better match user perceptions, and work [33] set performance benchmarks such as 90% acceptability rates in negotiation systems. Onboarding strategies, according to [34], should center around key activation points, while the work presented in [37] encouraged CRM-style engagement, including intranet banners, to maintain user involvement.

In [22] the authors demonstrated that early onboarding metrics are strong predictors of long-term retention. The study achieved an 88.75% recall rate in predicting post-tutorial churn, underscoring the importance of monitoring early user behavior. Similarly, the work presented in [20] linked negative sentiment in support tickets to churn, advocating for sentiment analysis alongside usage logs to create a risk assessment framework. In another study [19], the authors found that low API or integration usage and infrequent support interactions are reliable churn signals, making these metrics essential for prioritizing retention efforts.

Other research [13,23] highlighted that selecting the right machine learning model is crucial for effective churn prediction highlighting the trade-offs between different approaches. For example, the authors in [13] compared decision trees and random forests, finding that decision trees excel in recall, making them ideal for minimizing missed churn risks, while random forests offer better F1-scores, which may be preferable when balancing precision and recall. Meanwhile, research [23] achieved near-perfect predictive performance with an AUC of approximately 0.99 but warned against neural networks due to their tendency to degrade over time without frequent retraining. On the operational side, authors in [16] highlighted the computational cost of XGBoost, noting that it produces 335% higher emissions than logistic regression, and recommended cloud-based solutions for scalability and efficiency.

Measuring the financial impact of churn ensures that retention efforts align with business objectives. The work presented in [38] demonstrated how CLV predictions can optimize budget allocation, ensuring marketing spending targets high-value users. Another key insight comes from [29], which found that organic search ads outperformed email campaigns in driving conversions, cautioning against overly aggressive retention tactics that may alienate users.

Finally, the importance of addressing computational demands and sustainability in SaaS systems was highlighted by studies [14,16,19,36]. Key recommendations included optimizing models for cloud deployment, minimizing carbon footprints, and ensuring scalable infrastructure for real-time, high-volume applications. For instance, study [14] pointed out the significant computational requirements of deep learning models, particularly for large firms. In addition, the authors in [19] focused on designing models suited for cloud-based SaaS environments, while in study [16] the authors evaluated the environmental impact, specifically the carbon cost, of model training. To support scalability, authors in [36] recommended hybrid computing approaches, especially for high-demand systems like billing.

5.2. Selection of Optimal Machine Learning Model for Decision-Making

Selecting the optimal machine learning model depends on the specific use case. The initial decision for a SaaS provider is to determine the primary business objective driving the need for machine learning. If the goal is the churn prediction, which is the most frequently cited use case, the provider should assess whether their dataset consists primarily of structured behavioral and transactional data. If the data is rich in clickstream logs, user sessions, or subscription data, ensemble models such as Random Forest or XGBoost stand out due to their strong performance in predictive accuracy and feature importance insights. These models are particularly effective when interpretability and scalability are important, especially in Business-to-Business (B2B) contexts where clear justification for churn predictions is essential.

However, in use cases where user behavior evolves over time, for instance, in freemium games or SaaS products with high session variability [23], hybrid architectures that integrate static and time-dependent features into LSTM networks (such as subscription age or payment history) should be considered. These models better capture temporal dynamics and early disengagement signals but at the cost of increased complexity and training resources. For SaaS vendors requiring real-time deployment or low-latency predictions, LightGBM offers a compelling balance between performance and computational efficiency and is especially suitable for streaming contexts or integration into cloud-based platforms.

For use cases focusing on user segmentation, where the goal is to group users into interpretable personas or strategic marketing groups, the nature of the input data plays a pivotal role. If the data includes demographic, firmographic, and behavioral attributes,

then traditional clustering methods like K-means are preferred for identifying meaningful segments, especially when enhanced with dimensionality reduction techniques such as UMAP or HDBSCAN to handle sparse datasets [26]. These approaches support exploratory analysis and are valuable when strategic alignment with qualitative insights from surveys or interviews is required.

For predicting CLV in B2B environments, where historical billing data, engagement metrics, and firmographic features such as company size or industry play a critical role, hierarchical ensemble models such as LightGBM were particularly effective. These models deliver strong predictive accuracy and can be paired with time-series techniques like Auto ARIMA to capture revenue trends across different customer segments [37]. Additionally, LASSO regression was applied in this context to enhance model interpretability and stability by selecting key predictive features in high-dimensional datasets. This approach allows SaaS providers to identify and prioritize high-value accounts, allocate marketing resources more efficiently, and optimize long-term revenue strategies.

If the focus is on user engagement, satisfaction, or loyalty, the decision is again based on the type of available data. For use cases involving UI/UX feedback, support ticket sentiment, or onboarding flows, vendors should consider deep learning models (like MLP or sentiment-enhanced LSTM) only if they have large datasets and need to identify subtle trends. For smaller datasets or when model explainability is prioritized, logistic regression or PLS-SEM models provide interpretable insights and can be used to connect interface design or feature adoption to engagement metrics.

When the business objective revolves around dynamic pricing or trial optimization, where scenario testing and profitability forecasting are key, models must support what-if simulations and business impact metrics. In these cases, gradient boosting techniques (XGBoost, GBM), or LASSO regression are optimal due to their ability to model nonlinear relationships and provide stable coefficients for strategic levers like trial duration or usage-based pricing. SaaS platforms featuring dynamic pricing models may also benefit from reinforcement learning to adjust prices in response to user behavior and network effects over time.

Finally, for the choice of the appropriate machine learning model, deployment and sustainability constraints should be considered. For SaaS vendors with limited engineering capacity or requiring real-time predictions integrated into CRM systems, models like Random Forest, XGBoost, or LightGBM can be deployed via REST APIs or embedded in cloud ML pipelines (as in AWS SageMaker). Additionally, in energy-sensitive environments, vendors should prefer computationally lightweight models and monitor carbon emissions, as studies showed large environmental variances between architectures.

Although the primary focus of the reviewed papers was on the performance and utility of data-driven decision support systems in SaaS, interpretability was also a key consideration in several of the examined studies. For instance, paper [15] employed Logistic Regression, a model valued for its transparency, and visualized its coefficients to help stakeholders understand how different usage factors affect churn predictions. Likewise, the paper [16] emphasized explainability by analyzing feature importance in Random Forest and Logistic Regression models to reveal key churn drivers, even when predictive performance was modest. Other studies such as [12,13,22] also used interpretable models (such as Decision Trees) or feature importance plots to support decision-making with understandable insights.

Figure 4 illustrates the recommended actions for selecting the optimal ML model based on the specific use case and context as derived from the results of the reviewed papers.

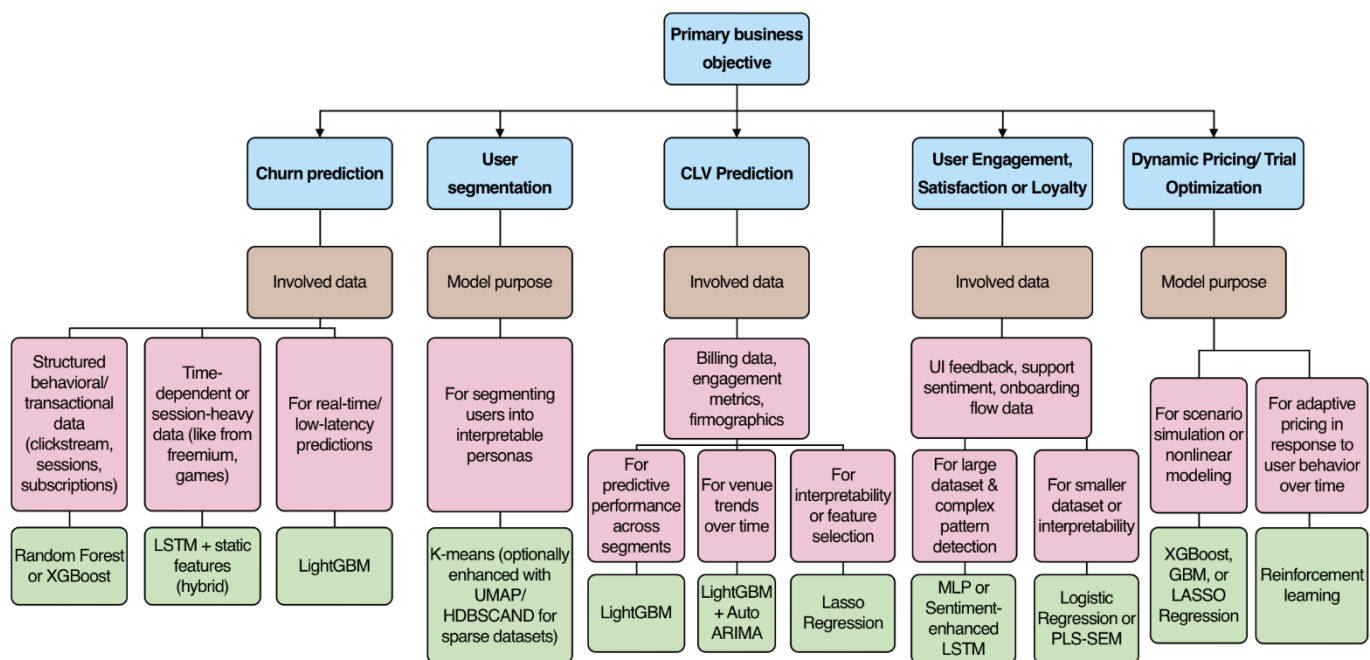


Figure 4. Taxonomy for ML algorithms selection for SaaS providers.

However, explainability techniques such as SHAP (SHapley Additive exPlanations) [55] or LIME (Local Interpretable Model-agnostic Explanations) [56] were not employed by the reviewed papers. SHAP is a method used to explain the output of any machine learning model by quantifying the contribution of each input feature to a specific prediction. It utilizes the concept of Shapley values from game theory to determine the importance of each feature. LIME is an interpretation technique used to explain the predictions of any black box machine learning model. LIME approximates a complex model's prediction with a simpler, interpretable model, focusing on individual predictions rather than the model as a whole. The choice between SHAP and LIME depends on the specific needs of the task. SHAP is ideal when accuracy, consistency, and fairness are priorities, especially in regulated settings or with tree-based models like XGBoost. LIME, on the other hand, is useful for quick, model-agnostic explanations during development, and works well with various data types. Its simplicity makes it great for explaining results to non-technical users. Ultimately, the adoption of such techniques could significantly improve stakeholder confidence in automated decisions and support error diagnosis, bias detection, and regulatory compliance.

Moreover, explicit considerations of algorithmic fairness, privacy protection, or ethical implications seem to be largely absent. Future research should address those gaps by incorporating such explainability techniques into machine learning models and consider the incorporation of privacy-preserving techniques such as differential privacy [57], federated learning [58] and secure multi-party computation, which can enable model training on sensitive user data without compromising user confidentiality. Additionally, incorporating algorithmic fairness assessments into the development pipeline could prevent discriminatory outcomes such as biased churn predictions. As SaaS solutions increasingly influence strategic decisions like marketing, pricing, or retention investments, it is critical to ensure that automated recommendations do not reinforce inequalities or introduce unfair treatment.

Even though carbon footprint should be taken into consideration, only two studies [13,15] measured how machine learning model selection affects carbon emissions. Study [15] examined the environmental impact of machine learning in B2B churn predic-

tion, analyzing the trade-offs between model accuracy and carbon emissions. Using an AMD EPYC 7763 processor (280 W) and European emissions data (296.96 g CO₂/kWh), the research quantified CO₂eq emissions in relatable terms. The results revealed that even basic preprocessing, such as feature creation, emits 86.379 g CO₂eq, while model training introduced far greater costs. The inclusion of multiple usage features resulting in a high dimensional dataset, significantly affected the carbon footprint produced by the reviewed machine learning methods. Specifically, Decision Trees and Random Forests showed a ~200% rise in emissions (DT: 2.349 g CO₂eq to 4.044 g; RF: 1.350 g to 4.044 g). XGBoost's emissions rose by 335% (21.29 g CO₂eq), and SVM increased by 50% (40.99 g CO₂eq), whereas Logistic Regression remained efficient (2.349 g CO₂eq).

However, the most striking environmental costs seemed to derive from deep learning and complex ensemble methods. In study [13], a Deep Neural Network (DNN) with two hidden layers (32/64 neurons) achieved a high accuracy percentage but at a high computational expense, scaling considerably as model complexness increased. Similarly, Random Forest proved resource-intensive, with costs growing alongside tree depth and feature complexity. Both DNN and RF required parallel computation to train efficiently and used up all of the available processors indicating their high resource consumption in comparison with the rest of the reviewed models. Although advanced models seemed to improve prediction, their energy demands make them unsustainable for firms without optimized infrastructure. The study concluded that businesses, especially smaller enterprises, should prioritize energy-efficient algorithms (like Logistic Regression) or cloud-based solutions to reduce environmental impact.

Although several of the reviewed studies experimented with deployment aspects, such as standalone real-time applications [5], CRM-integrated dashboards for decision support [14], cloud-based prediction APIs [29], and real-time data streaming using platforms like AWS SageMaker [20], these efforts primarily emphasize centralized, cloud-centric architectures. An emerging direction for real-world deployment of decision support systems in SaaS would involve combining centralized model training with decentralized inference through edge computing [59], particularly in settings involving IoT devices or local clients. Following this, machine learning models are trained in the cloud, often via MLaaS platforms that provide scalable computing resources, versioning and monitoring, and then deployed to edge devices for local inference. This approach could potentially offer key benefits such as reduced latency, improved bandwidth efficiency, offline functionality, and enhanced privacy, as predictions can be made without transmitting sensitive data back to the cloud.

6. Conclusions and Future Work

This work provided a scoping review of data-driven decision support systems (DSSs) within SaaS environments, highlighting current advances, practices, and gaps. Through an extensive analysis of recent literature, it has become clear that early churn prediction, customer segmentation, and personalized engagement strategies are critical for sustaining growth and profitability in SaaS businesses. Churn reduction emerges as the predominant research objective, with studies focusing on the identification of at-risk users and proposing targeted, data-driven interventions. From a methodological perspective, it was found that Random Forest and XGBoost algorithms consistently outperformed other machine learning models in churn prediction tasks, particularly when dealing with tabular behavioral and transactional data. Additionally, hybrid models combining static and sequential features have shown superior predictive performance, suggesting that integrating diverse data sources can enhance model robustness. However, simpler models like Logistic Re-

gression remain valuable for their interpretability, particularly in resource-constrained or real-time applications.

Beyond predictive modeling, research emphasized the importance of user segmentation and persona modeling for creating personalized engagement strategies. Key drivers of long-term retention identified across studies include UI/UX quality, effective onboarding (such as “aha moments”), flexible pricing strategies, and proactive support interventions. Moreover, linking predictive outputs to business impact metrics such as CLV, Monthly Recurring Revenue (MRR), and retention-related ROI have proven essential for bridging analytics with executive decision-making.

In terms of practical applications, many studies advocated for real-time deployment of predictive models through CRM integrations, APIs, and dashboard visualizations. This operationalization ensures that insights lead to immediate actions, supporting dynamic marketing, retention, and customer success initiatives. Moreover, choosing the right machine learning model for a SaaS provider depends on the business goal and data type. For churn prediction, ensemble models like Random Forest and XGBoost are ideal for structured behavioral data, offering accuracy and interpretability. When user behavior changes over time, LSTM-based models are better suited, though more complex. For real-time use, LightGBM balances speed and performance. In user segmentation use cases, models like K-means work well when combined with dimensionality reduction techniques, especially with diverse user data. For CLV predictions, LightGBM and LASSO regression provide accurate and interpretable insights, particularly when paired with time-series models. Regarding user engagement and satisfaction analysis, deep learning models are useful for large datasets, while logistic regression or PLS-SEM offer clarity with smaller ones. In dynamic pricing, XGBoost or reinforcement learning supports strategy testing and adaptive pricing. Finally, deployment requirements and energy efficiency are critical considerations. Models such as LightGBM, XGBoost, and Random Forest are not only known for their strong performance but are also relatively straightforward to deploy via REST APIs or integrate into cloud-based machine learning pipelines. However, in resource-constrained or energy-sensitive settings, where computational load and sustainability are priorities, lighter-weight models, such as logistic regression or decision trees should be prioritized to reduce infrastructure costs and minimize environmental impact without sacrificing essential performance.

However, despite the considerable progress in research, several challenges and open research directions persist. Machine learning models may exhibit performance degradation over time or struggle to adapt to evolving user behavior, highlighting the need for dynamic model monitoring and continual learning mechanisms. In order to ensure that developed DSSs remain effective, monitoring mechanisms should be established. Ensuring training data is up to date and machine learning models maintain good accuracy levels is crucial. Batch training of machine learning models may be a good starting point but will not be sustainable in the long term as data volume increases. To this end, periodically batch training of the models and/or real-time training incrementally updating the models could be explored in order to ensure data integrity. Therefore, MLaaS solutions [60] specifically designed for supporting decision-making in SaaS are of great importance since they can provide monitoring and updating mechanisms ready to use assuring the efficiency of DSSs.

Meanwhile, the environmental impact of machine learning implementations emerged as another critical consideration. Comparative analysis revealed that complex ensemble methods and deep neural networks can incur higher computational costs and carbon emissions than simpler model alternatives, underscoring the importance of balancing predictive performance with sustainability, particularly for SaaS companies managing significant computational workloads. Future research, except for seeking greener architectures, could

also emphasize clearly explaining the trade-offs between accuracy metrics and environmental footprint. Informing users of energy consumption and carbon emissions can create awareness and lead to more environmentally friendly model selection decisions like smaller architectures and the use of techniques like early stopping and transfer learning.

As decision-making increasingly relies on automated systems, ensuring the explainability, fairness, and transparency of predictions, particularly in churn risk assessments, will be critical. SaaS vendors need to have a clear view of what is happening in their business at any time and act towards improving user experience and ultimately increasing KPIs that are vital for their business. Therefore, it's important that DSSs justify their recommended decisions and provide clear guidance on what actions should be taken. Towards this direction, DSSs can utilize interpretable machine learning models to justify their decision-making processes. Post hoc methods for non-interpretable models such as SHAP values [55] and LIME [56] can be used to provide explanations on models' predictions allowing vendors to understand what influences the decisions. While both SHAP and LIME offer valuable insights, they serve distinct purposes. For SaaS providers, SHAP may be preferable for strategic decision-making and bias detection, while LIME could better support customer-facing teams needing immediate, case-specific explanations.

Natural language UI could also be explored to provide an easier and friendlier way to provide guidance to SaaS vendors. Large Language Models (LLMs) could be employed to understand user's intent and explain recommended actions, for example for reducing churn or increasing user engagement. Complicated tools and UIs are not the best choice for supporting SaaS professionals. They need to stay ahead of the competition and act fast and efficiently. Providing clear explanations in natural language and suggesting a set of actions to perform could adjust better to busy schedules and offer clarity.

Additionally, future systems may benefit from refined personalization strategies driven by context-aware recommendations, dynamically adapting to each user's lifecycle stage and dynamic profile. In particular, the DSSs would recommend a personalized course of actions tailored to each user's holistic profile which will be dynamically updated. Ultimately, the development of intelligent, adaptive, and sustainable decision support systems will be a key differentiator for SaaS providers seeking to build resilient, customer-centric businesses in an increasingly competitive and dynamic marketplace.

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